

Genetic report · longevity

Brain and Neurodegeneration

APOE, Alzheimer's, Parkinson's, ELA/DFT, cerebrovascular disease and white matter

Sérgio F · Report date: 02-10-2026

Local DNA analysis (chip + high-confidence genomic imputation) with calibrated polygenic scores in the reference population. Conclusions based only on genes, divided into favourable and unfavourable and linked to 84 aging mechanisms. It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

Contents

• Summary	3
• Favourable and unfavourable	4
• APOE	5
• Alzheimer's disease (loci beyond APOE).....	6
• Parkinson	8
• ELA and frontal temporal dementia.....	9
• Cerebrovascular disease and white matter	10
• Other neurological diseases	11
• Polygenic scores (all).....	12
• Mechanisms of ageing.....	13
• Practical part: suggestions	14
• Limitations of array and analysis.....	15
• Methods, chain and sources	16

Summary

43

Curated SNPs

38

in the chip

4

imputed
($DR2 \geq 0.8$)

1

without chip
genotype/
imputation

10

Polygenic scores

This report covers genes and scores with more influence on brain aging: **APOE**, Alzheimer's loci (microglia, endocytosis, lipids), Parkinson's disease (LRRK2, GBA1, SNCA, MAPT), ELA and frontotemporal dementia (UNC13A, C9orf72, TMEM106B, GRN), cerebrovascular disease and white matter lesions.

In 4 lines: (1) **APOE $\epsilon 3/\epsilon 3$** , without $\epsilon 4$ or $\epsilon 2$: the most important individual factor is at the reference risk level; (2) Parkinson with low polygenic risk (8 and 35 percentiles), without LRRK2 G2019S or GBA1 variants, and **MAPT H2/H2** (protector, ~4% of people); (3) the polygenic background of Alzheimer without APOE is medium to slightly high (52–91 percentiles according to the score); (4) stroke with a low score (3.5 and 19 percentiles), but white matter lesions have a high score (the 92nd percentile), with 9p21 G/G and PMF1 C/C: the health of small vessels and sleep is the priority brain.

Polygenic scores update (03-10-2026). The scores were recalculated with more published recent scores (the same method and the same reference population). This box prevails over the previous values cited in the text.

- Alzheimer's disease: the new genomics give p75 and p85 (before p52); with APOE $\epsilon 3/\epsilon 3$, reading goes from medium to medium-high (genomic consensus p75). The suggestions remain.

Favourable and unfavourable

Favourable

- **APOE ε3/ε3** (rs429358 T/T imputed DR2 0.99; rs7412 C/C chip): without ε4.
- **Parkinson with low polygenic background:** percentile 7.7 (1805 variants) and 35 (90 variants); LRRK2 G2019S, GBA1 N370S/E326K/T369M absent; SNCA rs356182 and rs356219 without risk alleles.
- **MAPT H2/H2** (rs1052553 G/G, rs8070723 G/G, rs17649553 T/T): Parkinson's protector and progressive supranuclear paralysis; less tau.
- **Stroke with low score:** percentile 3.5 (metaPRS) and 19; HDAC9 and PITX2 (atrial fibrillation) without risk alleles.
- **No strong risk variants:** TREM2 R47H and R62H, ABI3 S209F and C9orf72 (risk haplotype) absent; TOMM40, CR1 and FERMT2 without risk alleles.
- **Multiple sclerosis at the 24th percentile,** without HLA-DRB1*15:01.

Unfavourable

- **Alzheimer without APOE with medium to high background:** the 91st percentile in the 39 variant score and 68 in the 22; the genomic (the most complete) gives 52. They contribute MS4A6A A/A, EPHA1 G/G, SLC24A4 G/G, PTK2B C/C and 1 copy in BIN1, CLU, PICALM, CD33, CD2AP and ABCA7 (microglia and endocytosis).
- **White matter lesions at the 92nd percentile** + PMF1/SLC25A44 C/C + 9p21 G/G: is more likely to be more susceptible to small brain vessel disease.
- **Without rare protective alleles** PLCG2 P522R, APP A673T and KL-VS (Klotho): typical, but without this reserve.
- **Parkinson's, individual loci:** LRRK2 rs76904798 T/T (2 copies, ~1.15x each), BST1 A/A, GPNMB A/A, TMEM175 and MCCC1 heterozygotes (by lysosome), compensated by the low global score.
- **ELA/DFT:** UNC13A rs12608932 A/C (1 copy), GRN rs5848 C/T (less progranulin) and TMEM106B heterozygote.
- **Dementia (27 variants) at the 74th percentile** with low coverage (low reliability).

In Alzheimer's scores the percentile varies greatly with the number of variants. The genomic score (249 thousand variants, 82% of weighted coverage) is the most stable and gives the 52nd percentile; the scores of a few dozen variants overestimate the effect of each locus.

APOE

APOE is the major common genetic factor of Alzheimer's disease and cerebral vascular aging. The combination of rs429358 and rs7412 defines $\epsilon 2/\epsilon 3/\epsilon 4$. Sérgio F has **rs429358 T/T** (imputed, DR2 0.99; it is not on the chip) and **rs7412 C/C** (chip): **$\epsilon 3/\epsilon 3$** , the reference genotype (~60% of the population). It is consistent with previous reports. Rs4420638 (APOC1) A/G, which usually follows the $\epsilon 4$, here has no associated $\epsilon 4$: small effect.

Favourable (1)

Gene	Variant	rsID	Genotype	Source	Effect allele	Copies	Interpretation	Allele effect
APOE	APOE C112R (define $\epsilon 4$)	rs429358	TT	imputation DR2=0.99	C	0	favourable	Allele C = $\epsilon 4$: Alzheimer's ~3x (1copy) to ~12x (2copy) [strong]

Typical (without the rare protective allele) (1)

Gene	Variant	rsID	Genotype	Source	Effect allele	Copies	Interpretation	Allele effect
APOE	APOE R158C (define $\epsilon 2$)	rs7412	CC	chip	T (protector)	0	typical	Allele T = $\epsilon 2$: less Alzheimer's (~0.6x), but more triglycerides and dysbetalipoproteinemia in homozygous [strong]

Alzheimer's disease (loci beyond APOE)

Alzheimer's loci are concentrated in three pathways: microglia/innate immunity (TREM2, CD33, MS4A, PLCG2, ABI3, CR1), endocytosis and traffic of APP (BIN1, PICALM, SORL1, CD2AP) and lipids (ABCA7, CLU). Sérgio F has the common variants of small effect in the three pathways, without any rare variant of strong effect.

Polygenic scores of this section

Score	Trait	Percentage*	z	Interpretation	Variants used	Weighted coverage	Reliability	Method / publication
PGS000334	Alzheimer's (22 variants include APOE) §20	67 5 (above the average)	0 09	unfavourable	22 of 22	100%	high	Clumping and Thresholding (C+T); PGP000101, PMID 32968074
PGS000811	Alzheimer's (39 variants) §20	90 8 (high)	1 35	unfavourable	37 of 39	85%	high	Genome-wide signifiers; PGP000196, PMID 33532541
PGS002249	Alzheimer's (genomic 249 thousand variants) §20 profile; the highest coverage	52 3 (medium)	0 01	typical	212 109 of 249 273	82.2%	moderate	LD plumping; PGP000276, PMID 31302669
PGS003334	Dementia (27 variants) Low coverage	74 2 (above the average)	0 64	unfavourable	12 of 27	26.4%	low	Genome-wide signifier SNPs; PGP000402, PMID 36347157

* Percentage in the reference population (n=633), calculated with exactly the same variants that were used for Sérgio F (chip + imputation DR2 ≥ 0.8, dosages). A percentile is not an absolute risk: it indicates the relative position of the polygenic fundus.

Favourable (6)

Gene	Variant	rsID	Genotype	Source	Effect allele	Copies	Interpretation	Allele effect
TOMM40	TOMM40	rs2075650	AA	chip	G	0	favourable	G allele: in LD with ε4 [strong]
CR1	CR1	rs6656401	GG	chip	A	0	favourable	Allele A: Alzheimer ~1.2× (complement) [strong]
FERMT2	FERMT2	rs17125944	TT	chip	C	0	favourable	Allele C: Alzheimer ~1.1× [moderate]
ABI3	ABI3 S209F	rs616338	CC	chip	T	0	favourable	Allele T (rare): Alzheimer ~1.4× [strong]
TREM2	TREM2 R47H	rs75932628	CC	previous report (genotype imputed by the laboratory)	T	0	favourable	T allele (R47H, rare): Alzheimer ~2-4× [strong]
TREM2	TREM2 R62H	rs143332484	CC	chip	T	0	favourable	Allele T (R62H): Alzheimer ~1.5-2× [strong]

Unfavourable (11)

Gene	Variant	rsID	Genotype	Source	Effect allele	Copies	Interpretation	Allele effect
APOC1/APOE	APOC1 (tag ε4)	rs4420638	AG	imputation DR2=1.00	G	1	unfavourable	G allele: in LD with ε4; without ε4 the effect is modest [strong]
BIN1	BIN1	rs6733839	CT	chip	T	1	unfavourable	T allele: more tau/endocytosis; Alzheimer's ~1.2× (2.the strongest locus) [strong]

CLU	CLU	rs11136000	TC	chip	C	1	unfavourable	C allele: Alzheimer ~1.15× (clusterin, chaperone) [strong]
PICALM		rs3851179	TC	chip	C	1	unfavourable	C allele: Alzheimer ~1.1× (protective T ~0.9×) [strong]
MS4A6A	MS4A	rs983392	AA	chip	A	2	unfavourable	Allele A: Alzheimer ~1.1× (microglia) [strong]
CD33	CD33	rs3865444	CA	chip	C	1	unfavourable	C allele: Alzheimer ~1.05× (A protector ~0.95×) [strong]
CD2AP	CD2AP	rs10948363	AG	chip	G	1	unfavourable	G allele: Alzheimer ~1.1× [moderate]
EPHA1	EPHA1	rs11771145	GG	chip	G	2	unfavourable	G allele: Alzheimer ~1.05× (A protector) [moderate]
ABCA7	ABCA7	rs3764650	TG	chip	G	1	unfavourable	G allele: Alzheimer's ~1.2× (lipids, phagocytosis) [strong]
SLC24A4		rs10498633	GG	chip	G	2	unfavourable	G allele: Alzheimer ~1.1× [moderate]
PTK2B	PTK2B	rs28834970	CC	chip	C	2	unfavourable	Allele C: Alzheimer ~1.1× [moderate]

Typical (without the rare protective allele) (4)

Gene	Variant	rsID	Genotype	Source	Effect allele	Copies	Interpretation	Allele effect
SORL1		rs11218343	TT	chip	C (protector)	0	typical	C allele (minoritary, ~4%): Alzheimer's ~0.85× (protector); T/T : typical [strong]
PLCG2	PLCG2 P522R	rs72824905	CC	chip	G (protector)	0	typical	G allele (P522R, rare): Alzheimer's ~0.7× and longevity (microglia) [strong]
APP	APP A673T	rs63750847	CC	chip	T (protector)	0	typical	T allele (A673T, rare): less β-amyloid; protector [strong]
KL (Klotho)	KL-VS F352V	rs9536314	TT	chip	G (protector)	0	typical	KL-VS = G allele (~15%); T/T = typical [moderate]

Parkinson

The most frequent monogenic forms (LRRK2 G2019S, GBA1) are absent. The haplotype MAPT H2/H2 is the most favourable finding of this report. The unfavourable individual loci point to the lysosomal pathway (LRRK2, TMEM175, GPNMB, BST1), but the genomic score of 1805 variants, which integrates them, is at percentile 7.7.

Polygenic scores of this section

Score	Trait	Percentage*	z	Interpretation	Variants used	Weighted coverage	Reliability	Method / publication
PGS000902	Parkinson (90 variants Nalls 2019)	35 1 (below the average)	-0.40	favourable	87 of 90	95.5%	high	Genome-wide signifiers; PGP000235, PMID 31701892
PGS000903	Parkinson (1805 variants Nalls 2019) The largest coverage	7 7 (low)	-1.41	favourable	1 702 of 1 805	88.8%	high	Clumping and Thresholding (C+T); PGP000235, PMID 31701892

* Percentage in the reference population (n=633), calculated with exactly the same variants that were used for Sérgio F (chip + imputation DR2 ≥ 0.8, dosages). A percentile is not an absolute risk: it indicates the relative position of the polygenic fundus.

Favourable (7)

Gene	Variant	rsID	Genotype	Source	Effect allele	Copies	Interpretation	Allele effect
LRRK2	LRRK2 G2019S	rs34637584	GG	chip	A	0	favourable	Allele A (G2019S): Parkinson ~25-40% throughout life in patients [strong]
GBA1	GBA1 N370S	rs76763715	TT	chip	C	0	favourable	Allele C (N370S): Parkinson ~3-5x [strong]
GBA1	GBA1 E326K	rs2230288	CC	chip	T	0	favourable	T allele (E326K): Parkinson ~1.8x [strong]
GBA1	GBA1 T369M	rs75548401	GG	chip	A	0	favourable	Allele A (T369M): Parkinson ~1.3x [moderate]
SNCA		rs356182	AA	chip	G	0	favourable	G allele: Parkinson ~1.33x (the strongest locus SNCA) [strong]
SNCA		rs356219	AA	chip	G	0	favourable	Allele G: Parkinson ~1.29x [strong]
MAPT (H1/H2)	MAPT H1/ H2	rs1052553	GG	chip	A	0	favourable	Allele A : haplotype H1: Parkinson's disease and progressive supranuclear palsy; G = H2 protector [strong]

Unfavourable (5)

Gene	Variant	rsID	Genotype	Source	Effect allele	Copies	Interpretation	Allele effect
LRRK2	LRRK2 5'	rs76904798	TT	chip	T	2	unfavourable	T allele (~14%): Parkinson ~1.15x per copy [strong]
TMEM175		rs34311866	TC	chip	C	1	unfavourable	C allele: Parkinson ~1.23x and dementia with Lewy bodies [strong]
BST1		rs4698412	AA	chip	A	2	unfavourable	Allele A: Parkinson ~1.1x [moderate]
MCCC1		rs12637471	GA	chip	G	1	unfavourable	Allele G: Parkinson ~1.18x [moderate]
GPNMB		rs199347	AA	chip	A	2	unfavourable	Allele A: Parkinson ~1.1x [moderate]

ELA and frontal temporal dementia

C9orf72 expansion (the most common genetic cause of ELA/DFT) is not visible in a chip; the haplotype where it occurs (rs3849942) is absent (C/C), which makes expansion less likely. UNC13A A/C: 1 copy of the allele that increases cryptic splicing when there is loss of TDP-43. RGN rs5848 C/T: slightly lower progranulin. TMEM106B heterozygous: 1 risk copy and 1 protective.

Favourable (1)

Gene	Variant	rsID	Genotype	Source	Effect allele	Copies	Interpretation	Allele effect
C9orf72	C9orf72 (haplotype)	rs3849942	CC	chip	T	0	favourable	T allele in strand + (= "A" in the study): haplotype where C9orf72 expansion occurs; C/C : without the risk haplotype [moderate]

Unfavourable (2)

Gene	Variant	rsID	Genotype	Source	Effect allele	Copies	Interpretation	Allele effect
UNC13A	UNC13A	rs12608932	AC	chip	C	1	unfavourable	C allele: ELA ~1.2 and cryptic splicing with loss of TDP-43 [strong]
GRN	GRN 3'UTR	rs5848	CT	chip	T	1	unfavourable	T allele: less progranulin; TT associated with FTD-TDP and hippocampal sclerosis [moderate]

Undefined effect allele (1)

Gene	Variant	rsID	Genotype	Source	Effect allele	Copies	Interpretation	Allele effect
TMEM106B		rs1990622	AG	imputation DR2=0.99	not defined	—	undefined	heterozygous: 1 copy of the risk allele (majority in the reference population) and 1 of the protector [strong]

Cerebrovascular disease and white matter

The stroke scores are low, but the white matter lesion volume score (small vessel disease, linked to cognitive decline and glymphatic drainage) is at the 92nd percentile. PMF1/SLC25A44 C/C (2 copies) and 9p21 G/G (2 copies) go in the same direction. 9p21 is mainly a coronary locus and aneurysms; in the brain it weighs on large vessels.

Polygenic scores of this section

Score	Trait	Percentage*	z	Interpretation	Variants used	Weighted coverage	Reliability	Method / publication
PGS000038	Stroke (90 variants)	19 1 (low)	-0.83	favourable	84 of 90	90.7%	high	Clumping and Thresholding (C+T); PGP000026, PMID 30355576
PGS002259	Stroke (metaPRS 534 variants) integrates vascular risk factors	3 5 (low)	-1.62	favourable	527 of 534	98.3%	high	metaPRS (elasticnet); PGP000285, PMID 34031205
PGS001534	Volume of white matter lesions (RM) marker of small vessel disease	92 3 (high)	1.51	unfavourable	301 of 355	83.8%	moderate	penalized regression; PGP000244, PMID 35324888

* Percentage in the reference population (n=633), calculated with exactly the same variants that were used for Sérgio F (chip + imputation DR2 $\geq$ 0.8, dosages). A percentile is not an absolute risk: it indicates the relative position of the polygenic fundus.

Favourable (2)

Gene	Variant	rsID	Genotype	Source	Effect allele	Copies	Interpretation	Allele effect
HDAC9	HDAC9/TWIST1	rs2107595	GG	chip	A	0	favourable	Allele A: stroke of large vessels $\sim 1.2\times$ [strong]
PITX2 (4q25)		rs6843082	AA	chip	G	0	favourable	G allele: atrial fibrillation $\sim 1.45\times$ [strong]

Unfavourable (2)

Gene	Variant	rsID	Genotype	Source	Effect allele	Copies	Interpretation	Allele effect
9p21 (CDKN2B-AS1)	9p21.3	rs10757278	GG	chip	G	2	unfavourable	G allele: coronary disease, intracranial aneurysms and stroke $\sim 1.2\text{--}1.3\times$ per copy [strong]
PMF1/SLC25A44		rs2984613	CC	imputation DR2=1.00	C	2	unfavourable	C allele: intracerebral hemorrhage $\sim 1.33\times$ and more white matter lesions [moderate]

Other neurological diseases

Multiple sclerosis at the 24th percentile and HLA-DRB1*15:01 negative (§18 profile). Narcolepsy: HLA-DQB1*06:02 negative (see sleep report). Huntington's disease, repeat ataxia and C9orf72 expansion are not testable with chip.

Polygenic scores of this section

Score	Trait	Percentage*	z	Interpretation	Variants used	Weighted coverage	Reliability	Method / publication
PGS000809	Multiple sclerosis §20	23 9 (below the average)	-0.71	favourable	101 of 127	84.4%	moderate	Clumping of variants associated with multiple sclerosis; PGP000194, PMID 34088990

* Percentage in the reference population (n=633), calculated with exactly the same variants that were used for Sérgio F (chip + imputation DR2 ≥ 0.8, dosages). A percentile is not an absolute risk: it indicates the relative position of the polygenic fundus.

Polygenic scores (all)

Score	Trait	Percentage*	z	Interpretation	Variants used	Weighted coverage	Reliability	Method / publication
PGS000334	Alzheimer's (22 variants include APOE) §20	67 5 (above the average)	0 09	unfavourable	22 of 22	100%	high	Clumping and Thresholding (C+T); PGP000101, PMID 32968074
PGS000811	Alzheimer's (39 variants) §20	90 8 (high)	1 35	unfavourable	37 of 39	85%	high	Genome-wide signifiers; PGP000196, PMID 33532541
PGS002249	Alzheimer's (genomic 249 thousand variants) §20 profile; the highest coverage	52 3 (medium)	0 01	typical	212 109 of 249 273	82.2%	moderate	LD plumping; PGP000276, PMID 31302669
PGS003334	Dementia (27 variants) Low coverage	74 2 (above the average)	0 64	unfavourable	12 of 27	26.4%	low	Genome-wide signifier SNPs; PGP000402, PMID 36347157
PGS000902	Parkinson (90 variants Nalls 2019)	35 1 (below the average)	-0 40	favourable	87 of 90	95.5%	high	Genome-wide signifiers; PGP000235, PMID 31701892
PGS000903	Parkinson (1805 variants Nalls 2019) The largest coverage	7 7 (low)	-1 41	favourable	1 702 of 1 805	88.8%	high	Clumping and Thresholding (C+T); PGP000235, PMID 31701892
PGS000038	Stroke (90 variants)	19 1 (low)	-0 83	favourable	84 of 90	90.7%	high	Clumping and Thresholding (C+T); PGP000026, PMID 30355576
PGS002259	Stroke (metaPRS 534 variants) integrates vascular risk factors	3 5 (low)	-1 62	favourable	527 of 534	98.3%	high	metaPRS (elasticnet); PGP000285, PMID 34031205
PGS001534	Volume of white matter lesions (RM) marker of small vessel disease	92 3 (high)	1 51	unfavourable	301 of 355	83.8%	moderate	penalized regression; PGP000244, PMID 35324888
PGS000809	Multiple sclerosis §20	23 9 (below the average)	-0 71	favourable	101 of 127	84.4%	moderate	Clumping of variants associated with multiple sclerosis; PGP000194, PMID 34088990

* Percentage in the reference population (n=633), calculated with exactly the same variants that were used for Sérgio F (chip + imputation DR2 $\geq$ 0.8, dosages). A percentile is not an absolute risk: it indicates the relative position of the polygenic fundus.

Mechanisms of ageing

Attachment of the findings to the mechanisms of objectives for immortality. Only the mechanisms with findings in this theme are listed.

Mechanism (from 84)	Genetic findings	Interpretation
Protein Aggregation	APOE ε3/ε3; MAPT H2/H2 (less tau); SNCA without risk; common Alzheimer's loci (BIN1, CLU, PICALM) with 1 copy	favourable
Lysosome dysfunction	GBA1 and LRRK2 G2019S absent; LRRK2 rs76904798 T/T, TMEM175, GPNMB, BST1, GRN (light risk)	unfavourable
Autophagy / Deregulated Macroautophagy	No strong variants in LRRK2/GBA1; TMEM106B heterozygous	typical
Neuroinflammation	Microglia: MS4A6A A/A, CD33 and ABCA7 heterozygotes; without TREM2 R47H; without PLCG2 P522R protector	unfavourable
Glymphatic dysfunction	PGS of white matter lesions at the 92nd percentile; PMF1 C/C	unfavourable
Hematoencephalic Barrier dysfunction	EPHA1 G/G; small vessel disease most likely	unfavourable
Vascular Aging	Stroke with a low score (3.5 and 19); HDAC9 and PITX2 without risk; 9p21 G/G	typical
Splicing Errors	UNC13A A/C (cryptic splicing with loss of TDP-43)	unfavourable
Mitochondrial dysfunction	TOMM40 rs2075650 A/A (without risk allele); MCCC1 heterozygous	favourable

Practical part: suggestions

All the lines below are suggestion (not applied). They are consistent with the v2 genetic plan and with the current protocol (pause curcumin, luteolin, apigenin and quercetin while ALT is high, as an unapplied suggestion; senolytic pulse only fisetin + quercetin and only with normal ALT; caution with telomerase activators; D3 10 000 IU; omega-3 3.2–4 g/day; ferritin below 100). To discuss with the doctor.

- **suggestion (not applied):** Regular aerobic exercise (≥ 150 min/week, with intervals) and strength training: it is the most evident intervention for small vessels, BDNF and microglia; even more important with BDNF Val/Met (cognition report).
- **suggestion (not applied):** Deep and regular sleep (glymphatic drainage, which removes β -amyloid and tau): see the sleep report; white matter at the 92nd percentile gives priority to this axis.
- **suggestion (not applied):** LDL and ApoB as low as possible (pitavastatin + ezetimibe + bempedoic acid; inclisiran to discuss, as in the v2 plan): protect large and small cerebral vessels.
- **suggestion (not applied):** Omega-3 3.2–4 g/day of EPA+DHA (with DHA ≥ 1 g), according to the current protocol: APOE $\epsilon 3/\epsilon 3$ responds well to DHA.
- **suggestion (not applied):** Lithium 5 mg: maintain; lithium in microdose is associated with less dementia in observational studies and inhibits GSK-3 β (tau).
- **suggestion (not applied):** Rapamycin 4 mg of 6 in 6 weeks: maintain (autophagy, more charged lysosomal pathway); retatrutide and dapagliflozin are associated with less dementia in observational studies.
- **suggestion (not applied):** Genetic homocysteine (MTHFR compound, §13 profile): maintain 5-MTHF 800 μ g and B12; high homocysteine is associated with cerebral atrophy and white matter lesions.
- **suggestion (not applied):** Avoiding Parkinson's risk exposures: pesticides (paraquat, rotenone), chlorinated solvents (trichloroethylene) and repeated cranial trauma.
- **suggestion (not applied):** Avoid chronic anticholinergics (diphenhydramine, hydroxyzine, oxybutynin) and benzodiazepines of daily use: they are associated with cognitive decline.
- **suggestion (not applied):** Protecting hearing (vision and hearing report): hearing loss is the major modifiable factor of dementia.
- **suggestion (not applied):** Curcumin, sometimes suggested for the brain, should be paused while ALT is high (suggestion, not applied; the real stack has not changed).

Limitations of array and analysis

- The C9orf72 expansion, the repetitions (Huntington, ataxias), the rare variants of APP/PSEN1/PSEN2 and CNVs are not testable with chip.
- APOE rs429358 is not on the chip; it was imputed with DR2 0.99 (high confidence) and coincides with previous reports.
- The TREM2 R47H is not on the chip and the local imputation has low confidence in this SNP; the C/C genotype of a previous report (imputed by the laboratory) was used.
- The PGS of white matter lesions comes from a model trained in a biobank (penalized regression); calibration in the reference population is empirical.
- The consumption chips genotype ~0.7 M positions; rare variants, indels, repeat expansions and CNVs are not reliable or do not exist in the data. An absence of a rare variant here does not exclude a pathogenic variant: only clinical sequencing (panel, exome or genome) does so.
- Statistical imputation (public panel of complete genomes to 30x) was only used with DR2 ≥ 0.8 ; below that SNP is treated as "SNP not found in the data".
- Polygenic scores indicate relative position in the reference population, explain a small fraction of risk and do not include lifestyle, medication or family history.

Methods, chain and sources

Data: chip genotypes (GRCh38, strand +) and local imputation with $DR2 \geq 0.8$; all analyzed in the local computer. No genotype was sent out; public databases (Ensembl/dbSNP, GWAS Catalog, PGS Catalog) only consulted public identifiers (rsIDs and IDs of scores).

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP. For each SNP the genotype was compared with the alleles of dbSNP: 43 concordant without conversion; 0 needed complement; 0 palindromic SNPs (A/T or C/G), solved by the strand + of the chip and by the allelic frequency. The risk alleles of the studies that were in the opposite strand were converted to the strand + (indicated in the SNP note).

Effect allele: defined by the literature and verified in GWAS Catalog (most significant association for the trait). "Favourable" : without the risk allele or with the protective allele; "unfavourable" : with 1–2 copies of the risk allele; "typical" : without a rare protective allele (as most people).

Polygenic scores: Harmonized coefficients of PGS Catalog (GRCh38). Score = Δ coefficient $\times$ dose of effect allele. Empirical percentile in the reference population ($n=633$) with the same variants. Reliability: high $\geq 85\%$ of weighted coverage, moderate 60–85%, low $<60\%$.

Count: 43 Curated SNPs: 38 chip, 4 imputed, 1 without chip genotype/imputation (1 with previous report genotype).

Main sources

- Bellengez C et al. New insights into the genetic etiology of Alzheimer's disease and related dementias. Nat Genet 2022.
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